
A Lifted Splitting Template for Convolutional Dictionary Updates: Unification, Channel Lifting, and Grouped Consensus

Anonymous submission

code, logs, and run configurations accompany this manuscript

Abstract

The dictionary update is the computational bottleneck of convolutional dictionary learning (CDL): its frequency-domain normal equations decompose into diagonal-plus-rank- K systems, where K is the number of training images, and the algorithms proposed for it — ADMM with an equality constraint solved by iterated Sherman–Morrison or conjugate gradient, ADMM consensus and its parallel and “3D” variants, spatial tiling, and accelerated proximal gradient (FISTA) — have been derived and implemented as separate methods. We show that all of them are instances of a single lifted, majorised splitting: partition the data-fidelity blocks among P dictionary copies coupled to one constraint variable, majorise the per-copy quadratic either exactly (a rank- r solve, r the blocks per copy) or by linearisation, and drive the coupling with an augmented-Lagrangian dual that may be switched off. One implementation of this iteration, instantiated only by parameter choices, reproduces the trajectories of the reference SPORCO solvers for all four families to relative deviations below 10^{-15} — machine precision, and bitwise equality for two of the four — on a fixed protocol. The template does more than re-derive known methods. Its partition parameter has an interior that the literature does not: *grouped consensus*, whose exact-solve work $\Theta(BrMN)$ and copy memory $2(B/r)MN$ interpolate the iterated Sherman–Morrison and consensus endpoints, whose optimal penalty we predict, and verify, to scale linearly with the group size, and whose measured serial cost is minimised at an interior group size — faster per iteration than both classical endpoints. For multi-channel signals the same iteration expresses both standard representations — a multi-channel dictionary with shared coefficient maps, and a shared single-channel dictionary with per-channel maps — and exposes an equivalence: lifting the shared-dictionary update over channels and taking a DFT along the channel axis yields a “channel-frequency” method whose support constraint mirrors the known 3D/consensus correspondence along the image axis; we prove this and confirm it numerically to 3.3×10^{-15} over 30 full CDL iterations against an independent dense implementation. Grouped cells run at practical scale: an image-grouped rank- C update for colour CDL matches the convergence of the standard folded consensus update while using $C \times$ fewer dictionary copies. All claims are validated against logged reference runs.

1 Introduction

Convolutional sparse coding (CSC) approximates a signal $\mathbf{s}$ by a sum of convolutions $\sum_m \mathbf{d}_m * \mathbf{x}_m$ between small filters $\mathbf{d}_m$ and sparse coefficient maps $\mathbf{x}_m$ [1, 2, 3]. Learning the filters from K training images — convolutional dictionary learning (CDL) — alternates sparse coding with a *dictionary update*, and it is the latter that dominates the cost and the algorithmic literature. In the DFT domain the update’s normal equations decompose per spatial frequency into systems that are

diagonal plus rank K [4], so the rank-one Sherman–Morrison identity that makes the sparse-coding step cheap [5] does not apply directly, and a family of workarounds has grown: solve the rank- K system iteratively by conjugate gradient (CG) [3]; solve it exactly by *iterated* Sherman–Morrison (ISM) [3]; restore rank one by spatially tiling the images into one large image [6]; give each image its own dictionary copy and couple the copies by ADMM consensus [6, 7], serially or in parallel; recast the ensemble as a three-dimensional volume (“3D”) [6]; or avoid linear solves entirely with an accelerated proximal-gradient (FISTA) update [8, 4]. The comparative study of Garcia-Cardona and Wohlberg [4] showed that these differently-derived methods are not as different as they look: the 3D method *is* the consensus method after a DFT along the image axis, and the performance ranking among the rest is explained by the structure of the linear system each one solves.

This paper takes that observation to its conclusion. If the known dictionary updates differ only in how they treat one shared structure, there should be a single derivation — one optimisation problem, one iteration — from which each of them follows by a choice of a few explicit parameters, and whose *other* parameter values are new algorithms. We construct such a template and use it in both directions.

Contributions.

1. **One iteration, four solvers (§3).** We pose the dictionary update as a lifted, majorised splitting: the data-fidelity blocks are partitioned among P dictionary copies tied to a single constraint variable; each copy minimises either its exact quadratic (a diagonal-plus-rank- r solve, r blocks per copy, by rank- r iterated Sherman–Morrison or CG) or a linearised majoriser (a gradient step); an augmented-Lagrangian dual couples copies to the constraint and may be switched off, in which case a Nesterov anchor turns the iteration into accelerated proximal gradient. A single implementation of this iteration, instantiated by parameter dictionaries only, reproduces the SPORCO reference implementations [10] of the ISM, CG, consensus, and FISTA updates *trajectory-for-trajectory* to relative deviations between 0 (bitwise) and 4×10^{-16} on a fixed protocol (§6.1); the tiling, 3D, and parallel-consensus variants are accounted for as an isometry, a change of basis, and an execution attribute of cells already in the lattice.
2. **The interior is not empty (§4, §6.2, §6.4).** Between the single-copy update (ISM/CG, rank K) and the fully split update (consensus, rank 1) lies *grouped consensus*: $P = K/r$ copies, each solving an exact rank- r system. Its solve work scales as $\Theta(KrMN)$ and its copy memory and transform work as $\Theta((K/r)MN)$, so the group size r is a dial between the ISM and consensus resource profiles — and because the two terms pull in opposite directions, measured serial time per iteration is U-shaped in r , with an interior group size beating *both* classical endpoints ($1.7\text{--}1.9\times$ at $K=16$). Extending the penalty-scaling argument of [4] predicts that the optimal coupling penalty grows linearly in r ; empirically the $\sigma \propto r$ rule makes convergence essentially independent of the partition while fixed- σ curves fan out. Setting the group size to its extremes reproduces the reference ISM and consensus runs to machine precision — the check that the interior cells are generated by the same iteration rather than by new code.
3. **Channel lifting and a new equivalence (§5, §6.3).** For C -channel data the template expresses both standard representations — multi-channel dictionary with shared maps, where channels multiply only the right-hand sides of one factorisation, and shared single-channel dictionary with per-channel maps, where channels multiply the rank — and both run at the reference protocol’s colour scale from the same implementation. Lifting the shared-dictionary update over channels and conjugating by a channel DFT yields a *channel-frequency* method whose constraint keeps only the channel-DC slice: the exact analogue, along the channel axis, of the 3D/consensus correspondence along the image axis (Proposition 1). An independent dense implementation of the channel-frequency form tracks the channel-lifted template instance to 3.3×10^{-15} over 30 CDL iterations. The correspondence also identifies the shared dictionary as a channel-constant multi-channel dictionary, tying the two representations together inside the lattice, and the image/channel duality of [4] becomes a transposition symmetry of the whole lattice. At colour scale, the image-grouped rank- C cell — a partition the reference library does not ship, since it folds channels onto the image axis — converges indistinguishably from folded consensus with $C\times$ fewer copies (§6.3).

Everything is validated against the reference implementation on logged, seeded runs; §6 states the protocol, and the accompanying logs contain every trajectory, converged dictionary, deviation, and

timing quoted here. We follow the study design of [4] exactly where the object is shared — the functional, constraint set, alternation and coupling, sparse-coding stage, data, and preprocessing are held fixed — because the claim is about the geometry of the dictionary update’s algorithm space, not about a new CDL formulation.

2 Problem setting

CDL is posed as [4]

$$\arg \min_{\{\mathbf{d}_m\}, \{\mathbf{x}_{k,m}\}} \frac{1}{2} \sum_k \left\| \sum_m \mathbf{d}_m * \mathbf{x}_{k,m} - \mathbf{s}_k \right\|_2^2 + \lambda \sum_{k,m} \|\mathbf{x}_{k,m}\|_1 \quad \text{s.t.} \quad \|\mathbf{d}_m\|_2 = 1 \quad \forall m, \quad (1)$$

with K training images $\mathbf{s}_k \in \mathbb{R}^N$ and M filters. The standard solver alternates one iteration of a convolutional basis-pursuit denoising (CBPDN) sparse-coding step with one iteration of a dictionary-update step, coupling the stages through their auxiliary variables [9]; we keep this structure, and the CBPDN stage itself (ADMM with rank-one frequency-domain solves [5]), fixed throughout.

The dictionary update treats the maps as fixed. Writing $X_{k,m}$ for the linear operator with $X_{k,m} \mathbf{d}_m = \mathbf{x}_{k,m} * \mathbf{d}_m$, $X_k = (X_{k,0} \ X_{k,1} \ \dots)$, and stacking the (zero-padded) filters into $\mathbf{d}$, the update is

$$\arg \min_{\mathbf{d}} \frac{1}{2} \sum_k \|X_k \mathbf{d} - \mathbf{s}_k\|_2^2 + \iota_{C_{\text{PN}}}(\mathbf{d}), \quad C_{\text{PN}} = \{\mathbf{d} : (I - PP^T)\mathbf{d} = 0, \|\mathbf{d}_m\|_2 = 1 \ \forall m\}, \quad (2)$$

where P restricts each filter to its support and the proximal map of $\iota_{C_{\text{PN}}}$ is support restriction followed by per-filter normalisation. In the DFT domain the normal equations of the quadratic part separate per spatial frequency into $M \times M$ systems with matrix $\sum_k \hat{\mathbf{x}}_k \hat{\mathbf{x}}_k^H + \sigma I$ — diagonal plus rank K : each training image contributes one rank-one term. Throughout, a *fidelity block* is one such term — one row block $(X_k, \mathbf{s}_k)$ of the least-squares operator — indexed by b in a block set $\mathcal{B}$ ($\mathcal{B} = \{1, \dots, K\}$ in the grayscale case; for multi-channel data §5 enlarges it).

3 The template: a lifted, majorised splitting

3.1 Lifted problem and master iteration

Choose a partition $\mathcal{P} = \{B_1, \dots, B_P\}$ of $\mathcal{B}$ and give each part its own dictionary copy:

$$\arg \min_{\mathbf{d}_1, \dots, \mathbf{d}_P, \mathbf{g}} \sum_{p=1}^P f_p(\mathbf{d}_p) + \iota_{C_{\text{PN}}}(\mathbf{g}) \quad \text{s.t.} \quad \mathbf{d}_p = \mathbf{g} \quad \forall p, \quad f_p(\mathbf{d}) = \frac{1}{2} \sum_{b \in B_p} \|A_b \mathbf{d} - \mathbf{s}_b\|_2^2. \quad (3)$$

On the scaled augmented Lagrangian with per-copy penalty σ , one iteration of the template performs, with a majoriser $Q_p(\cdot; \mathbf{y}_p)$ of f_p at an anchor $\mathbf{y}_p$,

$$\mathbf{d}_p^+ = \arg \min_{\mathbf{d}} Q_p(\mathbf{d}; \mathbf{y}_p) + \frac{\sigma}{2} \|\mathbf{d} - (\mathbf{g} - \mathbf{h}_p)\|_2^2 \quad (\text{per copy}) \quad (4)$$

$$\mathbf{g}^+ = \text{prox}_{\iota_{C_{\text{PN}}}} \left(\frac{1}{P} \sum_p (\alpha \mathbf{d}_p^+ + (1-\alpha)\mathbf{g} + \mathbf{h}_p) \right) \quad (\text{projected average}) \quad (5)$$

$$\mathbf{h}_p^+ = \mathbf{h}_p + \rho_d (\alpha \mathbf{d}_p^+ + (1-\alpha)\mathbf{g} - \mathbf{g}^+) \quad (\text{dual step}) \quad (6)$$

$$\mathbf{y}_p^+ = \mathbf{g}^+ + \frac{t-1}{t^+} (\mathbf{g}^+ - \mathbf{g}), \quad t^+ = \frac{1}{2} (1 + \sqrt{1 + 4t^2}) \quad (\text{anchor, if extrapolating}) \quad (7)$$

with relaxation α (equal to 1 in all runs reported here, matching the reference study’s fixed-parameter protocol). The template’s parameters are:

Partition $\mathcal{P}$. How the blocks are shared out: one copy ($P = 1$), one copy per block ($P = |\mathcal{B}|$), or grouped interiors (§4, §5).

Majoriser Q_p and inner solver. *Exact:* $Q_p = f_p$; per spatial frequency (4) is then $(\sum_{b \in B_p} \hat{\mathbf{x}}_b \hat{\mathbf{x}}_b^H + \sigma I) \hat{\mathbf{d}}_p = \sum_{b \in B_p} \hat{\mathbf{x}}_b^* \hat{\mathbf{s}}_b + \sigma(\hat{\mathbf{g}} - \hat{\mathbf{h}}_p)$, diagonal plus rank $r_p = |B_p|$, solved either exactly by rank- r_p iterated Sherman–Morrison (plain Sherman–Morrison when $r_p = 1$) or inexactly by warm-started CG. *Linearised:* $Q_p(\mathbf{d}; \mathbf{y}) = f_p(\mathbf{y}) + \langle \nabla f_p(\mathbf{y}), \mathbf{d} - \mathbf{y} \rangle + \frac{L}{2} \|\mathbf{d} - \mathbf{y}\|_2^2$, for which (4) is the closed form $\mathbf{d}_p^+ = (L\mathbf{y}_p - \nabla f_p(\mathbf{y}_p) + \sigma(\mathbf{g} - \mathbf{h}_p)) / (L + \sigma)$.

Table 1: The reference dictionary updates as template settings, and their measured agreement with the corresponding stock SPORCO solver runs (grayscale protocol of §6.1: identical inputs, seeds, penalties, budgets; deviations are the maximum per-iteration relative difference of the CDL functional and the relative Frobenius difference of the final dictionaries; 0 denotes bitwise-identical trajectories).

Reference solver	Partition	Majoriser / inner solve	$(\sigma, \rho_d, \text{anchor})$	Traj. dev.	Dict. dev.
ISM	$P=1$	exact, rank- K ISM	$(\sigma, 1, \text{no anchor})$	4.1×10^{-16}	1.2×10^{-15}
CG	$P=1$	exact, CG(10^{-3}), warm	$(\sigma, 1, \text{no anchor})$	2.0×10^{-16}	1.0×10^{-15}
Cns	$P=K$	exact, rank-1 SM	$(\sigma, 1, \text{no anchor})$	0	0
FISTA	$P=1$	linearised(L)	$(0, 0, \text{Nesterov})$	0	0
Tiled	$P=1$ after spatial-concatenation isometry (analysis only; no reference solver ships)				
3D	$P=K$ cell under copy-axis DFT (equivalence of [4]; analysis only)				
Cns-P	$P=K$ cell, distributed execution (validated through its serial form)				

Coupling (σ, ρ_d, α) **and anchor.** The ADMM regime is $\sigma > 0$, $\rho_d = 1$, anchor unused. Setting $\rho_d = 0$ and $\sigma = 0$ removes the dual coupling entirely; with the linearised majoriser and the Nesterov anchor (7), the iteration (4)–(7) is then exactly accelerated proximal gradient (FISTA) on (2). Intermediate settings are linearised ADMM [18] and, at $P > 1$, proximal-gradient consensus, adjacent to the accelerated-proximal-gradient consensus CDL of [14].

Basis. All computation lives in the spatial DFT domain, but the iteration is covariant under unitary changes of variables (Lemma 1); bases along the *copy* and *channel* axes generate the 3D-type methods (§3.2, §5).

Representation. How channels enter $\mathcal{B}$ and C_{PN} (§5).

One code path implements (4)–(7); every method in this paper is a parameter dictionary handed to that implementation, and the sparse-coding stage, warm starts, and functional evaluation are shared verbatim with the reference solvers. Two initialisation conventions are inherited from the reference implementation rather than chosen by us (its equality-constraint and consensus classes initialise the warm-started dual differently, $\mathbf{h}^0 = \mathbf{g}^0$ versus $\mathbf{h}_p^0 = \mathbf{g}^0 / \sigma$); the template adopts the matching convention at each endpoint and the consensus convention elsewhere.

Lemma 1 (Unitary covariance). *Let U be unitary on the lifted space, acting on the copies $(\mathbf{d}_1, \dots, \mathbf{d}_P)$ and duals, and let $\tilde{f} = f \circ U^{-1}$, $\tilde{E} = UE$ where $E\mathbf{g} = (\mathbf{g}, \dots, \mathbf{g})$ is the lifting map. Then the iterates of (4)–(6) for the transformed problem $(\tilde{f}, \tilde{E})$ are the U -images of the iterates for (f, E) , with the same $\mathbf{g}$ sequence.*

Proof. Each step minimises a sum of terms that are either composed with U^{-1} (the fidelity), invariant under U (the quadratic coupling $\|\mathbf{d} - E\mathbf{g} + \mathbf{h}\|_2^2 = \|U\mathbf{d} - \tilde{E}\mathbf{g} + U\mathbf{h}\|_2^2$), or functions of $\mathbf{g}$ alone; minimisation commutes with the unitary change of variables, and the dual update is linear. The $\mathbf{g}$ -step depends on the copies only through $E^T(\mathbf{d} + \mathbf{h}) = \tilde{E}^H(U\mathbf{d} + U\mathbf{h})$. $\square$

3.2 The known updates as parameter settings

Table 1 lists the settings that reproduce each solver family of the reference implementation. Three known methods are accounted for without their own cells. *Spatial tiling* [6] concatenates the K images into one large image: an isometric re-indexing of the block set into a single rank-one block on a larger spatial grid, i.e. the $P = 1$ cell after an embedding that changes N rather than the iteration. *Parallel consensus* (Cns-P) is the $P = K$ cell executed with its copies distributed; the projected average (5) is the only synchronisation point, an observation that extends to every partition (§4). The *3D method* is the $P = K$ cell conjugated by a DFT along the copy axis: the consensus constraint (all copies equal) becomes a support constraint (only the copy-DC slice may be nonzero) and the block-diagonal fidelity becomes block-circulant. This is the correspondence proved in [4]; in the template it is one instance of Lemma 1, and §5 instantiates the same mechanism along a different axis to obtain a new equivalence.

4 The interior of the partition parameter: grouped consensus

The two ADMM endpoints differ only in $\mathcal{P}$, so the partitions between them are algorithms too: split the K images into $P = K/r$ groups of r and solve each copy’s diagonal-plus-rank- r system exactly by rank- r iterated Sherman–Morrison. To our knowledge this *grouped consensus* has not been instantiated for the convolutional dictionary update; it is the specialisation to (2) of general-form consensus ADMM [7], and its properties follow from the linear-system structure:

Work. An exact rank- r solve costs r sequential rank-one Sherman–Morrison stages, each $\Theta(rMN)$ across frequencies, so the solve work per iteration is $\Theta(Pr^2MN) = \Theta(KrMN)$: linear interpolation (in r) between consensus $\Theta(KMN)$ and ISM $\Theta(K^2MN)$ — the $\mathcal{O}(K^2)$ against $\mathcal{O}(K)$ dichotomy of [4] becomes a slope. The *total* per-iteration cost adds the per-copy transforms and the projected average, $\Theta(PMN \log N) = \Theta((K/r)MN \log N)$, which *fall* as groups coarsen; the sum $c_1Kr + c_2(K/r)$ is therefore U-shaped in r , with the striking consequence — confirmed in §6.4 — that an interior group size minimises serial time per iteration, beating *both* classical endpoints.

Memory and parallelism. Copies and duals occupy $2PMN = 2(K/r)MN$, interpolating $2MN$ (ISM) and $2KMN$ (consensus); the copies solve independently with the single synchronisation (5), so r selects fewer, larger parallel workers — the natural cell when cores or memory are scarcer than K .

Penalty scaling. The reference study derives, by a data-replication argument, that the optimal σ grows as $\Theta(K)$ for the equality form and $\Theta(1)$ for consensus. The same argument applied per copy gives the general law: replicating a block r times multiplies the copy’s fidelity, but not its coupling term, by r , so equivalence across group sizes requires $\sigma \propto r$ — recovering both known endpoints ($r=K$ and $r=1$) and predicting the interior. §6.2 confirms it: under a fixed σ the convergence curves order themselves by P , while under $\sigma = \sigma_1 r$ they collapse onto each other, making the partition a resource dial with no convergence price.

5 Multi-channel geometry

For C -channel signals $\mathbf{s}_{c,k}$ there are two standard convolutional representations [11, 4], and the template expresses both through the block set and the constraint:

Multi-channel dictionary, shared maps ($\mathcal{R}_1$): filters $\mathbf{d}_{c,m}$, one map set $\mathbf{x}_{k,m}$ shared by all channels. The update’s fidelity $\frac{1}{2} \sum_{k,c} \|X_k \mathbf{d}_c - \mathbf{s}_{c,k}\|^2$ is block-diagonal over channels *with the same operator* X_k in every channel: per spatial frequency there are C right-hand sides for one diagonal-plus-rank- K system, so any grayscale cell applies unchanged with its factorisation shared across channels — the reason the multi-channel dictionary update costs no more than the grayscale one [4]. (Following the reference implementation, each filter is normalised jointly over its channels; the formulation in [4] writes the constraint per channel, a discrepancy we inherit from the software and disclose.)

Shared single-channel dictionary, per-channel maps ($\mathcal{R}_2$): one filter set $\mathbf{d}_m$, maps $\mathbf{x}_{c,k,m}$. Every (k, c) pair is now a fidelity block acting on the *same* $\mathbf{d}$: $\mathcal{B}$ has $B = CK$ blocks and the single-copy system has rank CK — channels multiply the rank, the dual role to $\mathcal{R}_1$ where they multiply right-hand sides. The reference implementation reaches this case by folding channels onto the image axis and running its grayscale solvers, i.e. it instantiates exactly the $P = 1$ and $P = CK$ partitions. The interior is new and now two-dimensional: group the (k, c) blocks *by image* ($P = K$ copies, exact rank- C solves) or *by channel* ($P = C$ copies, exact rank- K solves), or coarser groupings of either kind. A third representation — per-channel maps coupled by a joint-sparsity $\ell_{2,1}$ term [11] — alters only the sparse-coding stage; its dictionary update is identical to $\mathcal{R}_2$ ’s, so every cell here serves it unchanged.

The channel-grouped cells are not merely permitted by the template; one of them is forced into view by the same change of basis that linked 3D to consensus:

Proposition 1 (Channel-frequency form of the shared-dictionary update). *Let F_C be the unitary DFT along the channel axis, and consider the channel-lifted instance of (3) for $\mathcal{R}_2$: copies $\mathbf{d}_c$ with $f_c(\mathbf{d}) = \frac{1}{2} \sum_k \|X_{k,c} \mathbf{d} - \mathbf{s}_{c,k}\|^2$ and constraint $\mathbf{d}_c = \mathbf{g} \forall c$. Conjugating by F_C maps this instance, iterate for iterate (Lemma 1), to the channel-frequency method: a lifted variable $\tilde{\mathbf{d}} = F_C(\mathbf{d}_0, \dots, \mathbf{d}_{C-1})$ with (i) fidelity given by the block-circulant operator with symbols $T_{k,j} = \frac{1}{C} \sum_c \omega^{-jc} X_{k,c}$, $\omega = e^{2\pi i/C}$*

— a C -point convolution along channel frequency — and (ii) the constraint $\tilde{\mathbf{d}}_j = 0$ for $j \neq 0$, $\tilde{\mathbf{d}}_0 = \sqrt{C} \mathbf{g}$ with $\mathbf{g} \in C_{\text{PN}}$: a support condition selecting the channel-DC slice, of exactly the form that defines the 3D method along the image axis [6, 4]. Consequently (iii) the shared single-channel dictionary is the multi-channel dictionary constrained to be channel-constant, i.e. $\mathcal{R}_2$ is $\mathcal{R}_1$ plus a support constraint in the channel-frequency domain, and (iv) the channel-frequency method and channel-lifted consensus are the same algorithm in different coordinates.

Proof. The fidelity of the lifted instance is block-diagonal along channels; a diagonal operator along an axis is circulant along the conjugate (DFT) axis, with symbols the DFT of its diagonal blocks, which gives (i) with the unitary normalisation absorbed as stated. The lifting map $E\mathbf{g} = (\mathbf{g}, \dots, \mathbf{g})$ transforms as $F_C E \mathbf{g} = (\sqrt{C}\mathbf{g}, 0, \dots, 0)$, giving (ii). Applying Lemma 1 with $U = F_C$ yields (iv); (iii) is (ii) read as a statement about the constraint set of the multi-channel-dictionary representation. The dual initialisation transforms consistently: $\mathbf{h}_c^0 = \mathbf{g}^0/\sigma$ maps to $\tilde{\mathbf{h}}^0 = (\sqrt{C}\mathbf{g}^0/\sigma, 0, \dots, 0)$. $\square$

Proposition 1 is the channel-axis transposition of the image-axis correspondence in [4], and it makes the image/channel duality observed there — C controls the sparse-coding rank as K controls the update rank — act on the whole lattice of instances: exchanging the roles of k and c exchanges by-image with by-channel grouping, and the 3D method with the channel-frequency method. Because a proposition of this kind is only as credible as its bookkeeping, we also validate it operationally: an *independent* implementation of the channel-frequency method — dense $(CM) \times (CM)$ per-spatial-frequency solves on the block-circulant operator, no Sherman–Morrison, no return to channel space — run inside the full CDL loop, against the channel-lifted template instance (§6.3).

6 Experimental validation

Protocol. All experiments follow the reference study’s fixed-parameter protocol on MIRFLICKR-derived images [12] with its preprocessing (scaling to $[0, 1]$, optional grayscale conversion, Tikhonov highpass with regularisation 5.0); training images are selected, and initial dictionaries drawn, from a logged seed (12345). Both sides of every comparison run the stock CBPDN sparse-coding stage with one inner iteration per alternation, auxiliary-variable coupling, fixed penalties (no adaptation, no relaxation, no backtracking — the settings under which the reference study reports its main comparisons), and the same functional evaluation: the CDL objective (1) at the constraint-set dictionary iterate (unit-norm filters by construction; logged per-filter norms are 1 to machine precision) and the sparse stage’s auxiliary maps. Reference numbers are produced by running the stock SPORCO ConvBPDNDictLearn solvers at runtime on the same inputs; nothing is transcribed from the literature. Complete configurations, trajectories, converged dictionaries, and deviations for every run are in the accompanying logs.

6.1 Reproducing the four solver families

On the grayscale protocol ($K=5$ images at 64×64 , $M=16$ filters of 8×8 , $\lambda = 0.1$, 50 alternations; penalties from the reference study’s $K=5$ grid optima: $\rho = 3.59$, $\sigma = 4.08$ for ISM/CG, $\sigma = 1.29$ for consensus, $L = 48.14$ for FISTA), each template instance was run against the stock solver of its family on identical inputs. Table 1 and Figure 1 give the outcome: the consensus and FISTA instances are *bitwise identical* to their references at every logged iteration, and the ISM and CG instances agree to 4.1×10^{-16} and 2.0×10^{-16} respectively (double-precision roundoff; the final dictionaries agree to 1.2×10^{-15} and 1.0×10^{-15}). The agreement sits at double-precision roundoff, more than twelve orders of magnitude below the 10^{-3} level at which trajectories would count as matching; we take this as the operational demonstration that the four families are one iteration.

6.2 Grouped consensus: collapse checks, the $\sigma \propto r$ law

Grouped consensus was run at $K = 8$ (64×64 , $M = 16$, 50 alternations) for $P \in \{1, 2, 4, 8\}$. First the internal-consistency anchors: setting the group size to its extremes must, if the interior cells are generated by the same iteration, land exactly on the reference solvers. They do: $P=1$ (one group of 8) versus stock ISM at $\sigma = 11.0$ deviates by 3.8×10^{-16} (dictionaries 1.5×10^{-15}), and $P=8$ versus stock consensus at $\sigma = 2.2$ is bitwise identical. Second, the interior (Figure 2): at a common $\sigma = 4.0$

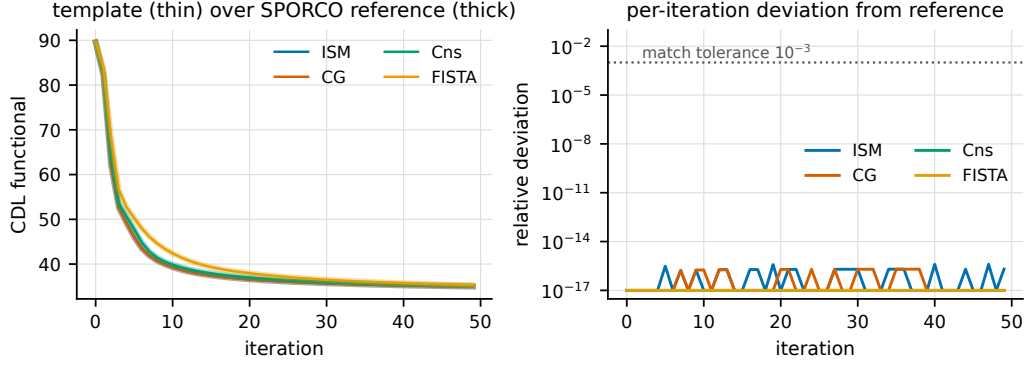


Figure 1: Template instances against stock SPORCO dictionary-update solvers, grayscale protocol of §6.1. Left: CDL functional per alternation; each thin template curve lies on its thick reference curve. Right: per-iteration relative deviation; consensus and FISTA are exactly 0 at every iteration (clipped to the plot floor), ISM and CG sit at double-precision roundoff. The dotted line marks the 10^{-3} level.

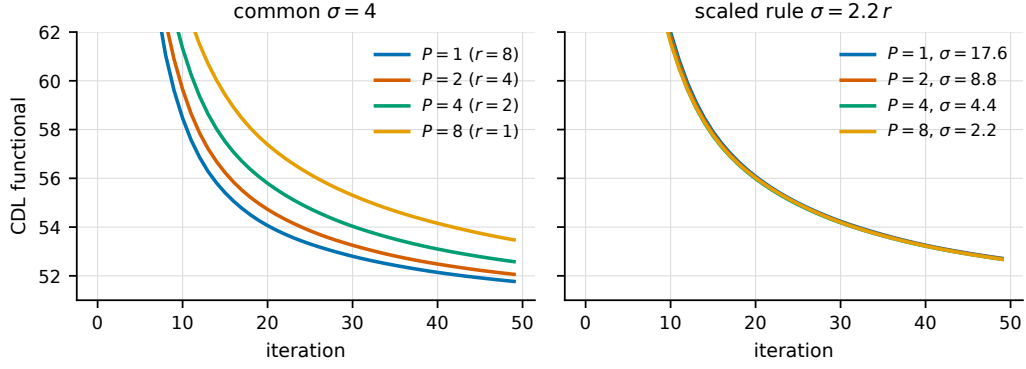


Figure 2: Grouped consensus, $K = 8$ grayscale (P copies of group size $r = 8/P$; $P=1$ is the ISM cell, $P=8$ the consensus cell). Left: fixed $\sigma = 4$; convergence orders by copy count. Right: the penalty-scaling rule $\sigma = 2.2 r$ predicted in §4 collapses the curves.

the final functional degrades monotonically with the number of copies (51.77 at $P=1$ to 53.47 at $P=8$) — more duals, more coupling lag per iteration — while under the predicted rule $\sigma = 2.2 r$ the four curves collapse (finals 52.72, 52.70, 52.68, 52.68; spread 0.036 against 1.70 at fixed σ). The partition is thus a genuine resource dial: group size trades solve work $\Theta(KrMN)$ and copy memory $2(K/r)MN$ with no per-iteration convergence cost once σ follows the scaling law.

6.3 Multi-channel instantiations and the channel-frequency equivalence

Multi-channel dictionary ($\mathcal{R}_1$). At the reference study’s colour scale — $K = 40$ colour images, 256×256 , $C = 3$, $M = 64$, $\lambda = 0.1$, 100 alternations — the template’s consensus cell against stock ConvBPDNDictLearn (cns, $\sigma = 2.2$, $\rho = 3.0$) and its FISTA cell against the stock PGM update ($L = 3 \cdot 14K = 1680$, $\rho = 2.2$, the stabilised step rule of the reference study’s colour experiment) again coincide to machine precision (trajectory deviations 3.3×10^{-16} and 0), so the grayscale-cell-with- C -right-hand-sides picture of §5 is exact in practice. Both runs converge properly as colour CDL: the functional falls from 21181.2 to 5562.3 (consensus cell; ratio 0.263) and to 5571.1 (FISTA cell) — far below the $0.6\times$ mark relative to the first-alternation value — and the learned filters are the expected oriented, opponent-colour atoms (Figure 4). Incidentally, the template’s vectorised copy update completed in 92 minutes against 111 for the stock per-block loop on this machine — an implementation observation, not an algorithmic claim, but a practical perk of the shared driver.

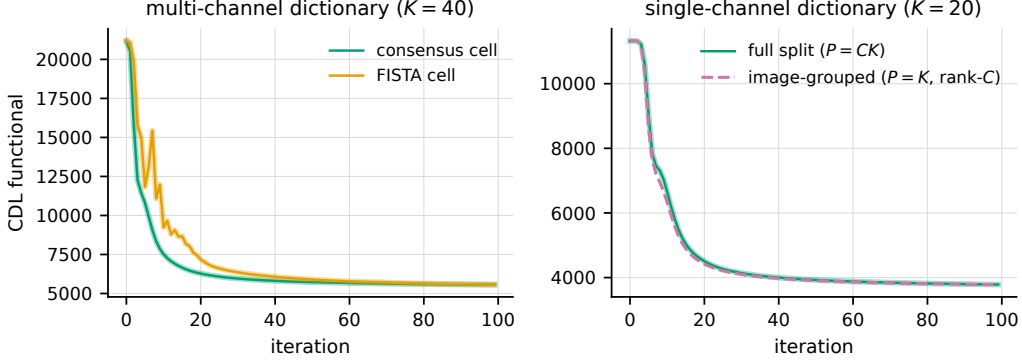


Figure 3: Colour CDL from the template, 256×256 MIRFLICKR images, $M = 64$, $\lambda = 0.1$. Left: multi-channel dictionary ($\mathcal{R}_1$, $K=40$): template consensus and FISTA cells over their stock references (thick; curves coincide). Right: shared single-channel dictionary ($\mathcal{R}_2$, $K=20$): the full-split cell over stock folded consensus, and the new image-grouped rank- C cell, which tracks the same convergence with $C \times$ fewer copies.

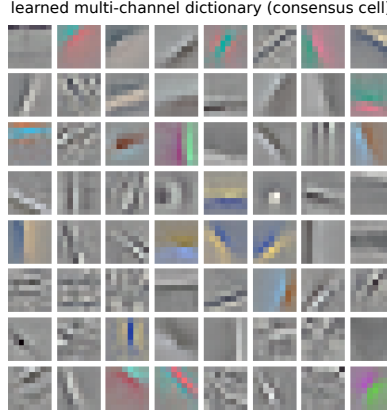


Figure 4: The $M = 64$ multi-channel 8×8 filters learned by the template's consensus cell at $K = 40$ (Figure 3, left).

Shared dictionary ($\mathcal{R}_2$). The plain per-channel-map representation (no joint-sparsity term, $\lambda = 0.1$) was run at $K = 20$, 256×256 — the reduced scale pre-declared in our plan when the $K=40$ folded problem's ≈ 50 GB working set exceeded this machine's memory; the $\mathcal{R}_1$ runs above are unaffected. The template's full-split cell against the stock folded-consensus reference is bitwise identical (deviation 0), and converges from 11323.2 to 3784.0 (ratio 0.334). The *image-grouped* cell — $P = K$ copies, each holding its image's C channel blocks and solving an exact rank- C system, a partition no shipped solver instantiates — reaches 3779.2 (ratio 0.334) on the same inputs: matching the folded reference's convergence, indeed marginally below it at this budget (Figure 3, right), with $C \times$ fewer copies and duals, i.e. the memory of grayscale consensus rather than of channel-folded consensus.

Partition cells at analysis scale. At $K = 5$, 64×64 , $C = 3$, $M = 16$, all four partitions of the (k, c) block lattice were run (Figure 5, left): the endpoints again land on stock ISM (2.5×10^{-16}) and stock folded consensus (bitwise), and the two grouped cells interpolate between them in copy count and in per-iteration convergence ($P=1$: 109.12; $P=C$: 109.53; $P=K$: 110.36; $P=CK$: 112.48 at 30 alternations), the same monotone-in- P ordering as in §6.2.

The equivalence, operationally. The channel-frequency method of Proposition 1 was implemented from its own definition — per spatial frequency, dense $(CM) \times (CM)$ normal solves on the block-

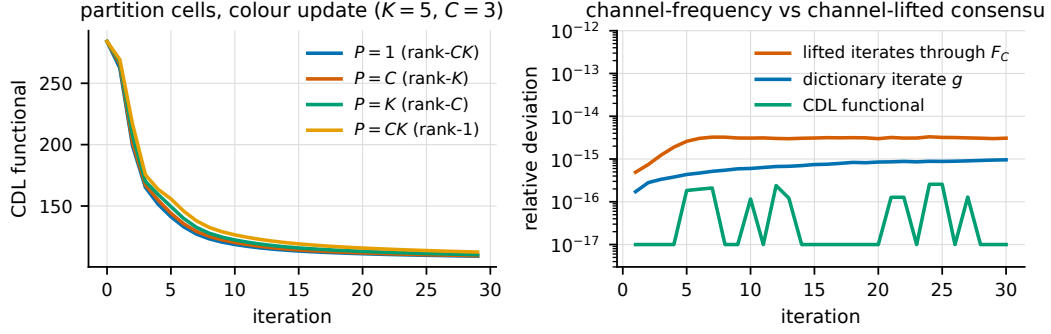


Figure 5: Left: the four partitions of the shared-dictionary colour update ($K=5, C=3$): endpoints match stock solvers, grouped cells interpolate. Right: iterate agreement between the independently implemented channel-frequency method and channel-lifted consensus across a full CDL run; all three traces sit at double-precision roundoff (Proposition 1).

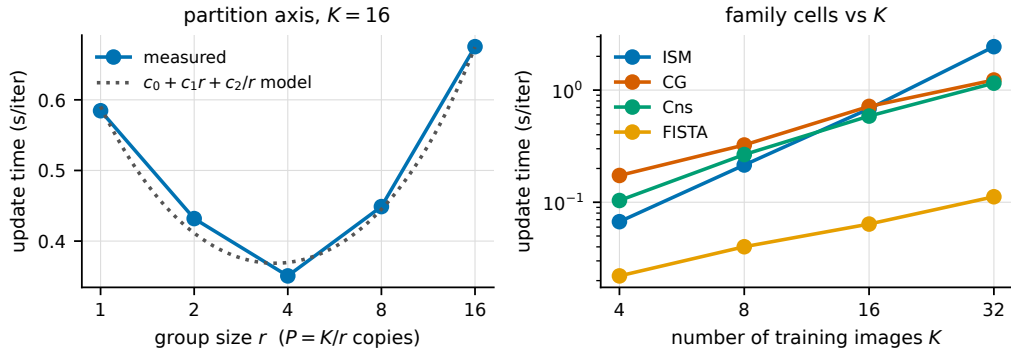


Figure 6: Measured dictionary-update time per iteration (single thread). Left: across group size r at $K=16$; the two-term cost model (solve work $\propto r$ plus per-copy transforms $\propto 1/r$) fits the U-shape, whose minimum beats both classical endpoints. Right: across K for the four family cells.

circulant operator with symbols $T_{k,j}$, the lifted variable kept in channel-frequency coordinates throughout — and run in the full CDL loop against the channel-lifted template instance ($P = C$, exact rank- K solves) on identical inputs. Over 30 alternations the lifted iterates agree through F_C to 3.3×10^{-15} , the dictionary iterates to 9.6×10^{-16} , and the functionals to 2.6×10^{-16} (Figure 5, right): the two methods are the same algorithm in different coordinates, as claimed, and the correspondence survives 30 rounds of interleaving with the sparse-coding stage.

6.4 Cost structure

Figure 6 tests the cost model of §4 on measured update times (single-threaded, 128×128 , $M = 32$). Across the partition at $K = 16$ the per-iteration time is U-shaped exactly as the two-term account predicts — 0.585 s at $r=1$ (consensus: many copies, many transforms), 0.351 s at $r=4$, 0.675 s at $r=16$ (ISM: quadratic solve work) — and the fit $c_0 + c_1 r + c_2/r$ tracks the measurements. We had anticipated only the solve term when planning these experiments; the measured minimum at an interior r is a sharper result than the monotone trade we expected, and it means grouped consensus is not merely a compromise between ISM and consensus but the serially fastest exact-ADMM update at this scale ($1.7\times$ faster per iteration than consensus, $1.9\times$ than ISM), while the $\sigma \propto r$ rule keeps its per-iteration convergence at parity. Across K the family cells reproduce the reference study's asymptotics — consensus and FISTA linear, ISM quadratic (2.440 s at $K=32$ against 1.154 s for consensus), CG in between with its tolerance-dependent constant — now as sections of one surface rather than as separate implementations.

Table 2: Validated template instances. “Agreement” is the maximum per-iteration relative deviation of the CDL functional from the stock SPORCO run on identical inputs (its paired internal-consistency run for cells with no shipped counterpart); “functional” is the converged CDL objective (1) at the stated budget. Full configurations and trajectories in the logs.

Method (instance)	Representation	Template setting	Agreement vs. reference	Functional
ISM	grayscale	$P=1$, exact rank- K , $(\sigma, 1)$	4.1×10^{-16} (stock ISM)	34.947
CG	grayscale	$P=1$, exact CG(10^{-3}), $(\sigma, 1)$	2.0×10^{-16} (stock CG)	34.950
Consensus	grayscale	$P=K$, exact rank-1, $(\sigma, 1)$	0 (stock Cns)	35.145
FISTA	grayscale	$P=1$, linearised, $(0, 0, \text{Nesterov})$	0 (stock PGM)	35.294
Grouped consensus $P=2$	grayscale, $K=8$	groups of 4, exact rank-4	endpoints: $3.8 \times 10^{-16} / 0$	52.70
Grouped consensus $P=4$	grayscale, $K=8$	groups of 2, exact rank-2	(vs stock ISM / stock Cns)	52.68
Consensus cell	colour $\mathcal{R}_1$, $K=40$	$P=K$, rank-1, C RHS	3.3×10^{-16} (stock Cns)	5562.3
FISTA cell	colour $\mathcal{R}_1$, $K=40$	linearised, $(0, 0, \text{Nesterov})$	0 (stock PGM)	5571.1
Full split	colour $\mathcal{R}_2$, $K=20$	$P=CK$, rank-1	0 (stock folded Cns)	3784.0
Image-grouped (new)	colour $\mathcal{R}_2$, $K=20$	$P=K$, exact rank- C	collapse: $2.5 \times 10^{-16} / 0$	3779.2
Channel-lifted (new)	colour $\mathcal{R}_2$, $K=5$	$P=C$, exact rank- K	3.3×10^{-15} (channel-freq. form)	109.53
Channel-frequency (new)	colour $\mathcal{R}_2$, $K=5$	channel-DFT basis, dense solve	3.3×10^{-15} (channel-lifted cell)	109.53

Table 3: The lattice of instances. $\checkmark$: reproduced against a stock solver at machine precision; $\equiv$: accounted for as a change of basis / isometry / execution mode of a validated cell; $\star$: new cell, run and anchored by its endpoint collapses; $\circ$: expressible, analysed, not run ($\dagger$: adjacent published method).

Partition of blocks	Exact (rank- r ISM)	Exact (CG)	Linearised
single copy ($r=B$)	ISM $\checkmark$; Tiled $\equiv$	CG $\checkmark$	FISTA $\checkmark$
grouped ($1 < r < B$)	gray $\star$; by-image $\star$; by-channel $\star$	$\circ$	$\circ$
full split ($r=1$)	Cns $\checkmark$; Cns-P, 3D $\equiv$	$\circ$ (rank-1 exact is closed form)	$\circ$ ($\dagger$ [14])
channel-DFT basis	channel-frequency $\star$ ($\equiv$ by-channel, Prop. 1)	(the basis change composes with either column)	

6.5 Summary of instances

Table 2 collects the validated instances; Table 3 maps the lattice.

7 Choosing a cell: mechanism and regimes

The lattice turns method selection into resource arithmetic, because every cost term is explicit in (B, r, M, N) with $B = K$ (grayscale, $\mathcal{R}_1$) or $B = CK$ ($\mathcal{R}_2$): exact-solve work $\Theta(BrMN)$, per-copy transform and averaging work $\Theta((B/r)MN \log N)$, copy/dual memory $2(B/r)MN$, one projected-average synchronisation per iteration, and — once $\sigma \propto r$ — per-iteration convergence independent of the partition. In serial the two work terms give an optimal interior group size $r^* \approx \sqrt{c_2/c_1}$ (empirically $r^* \approx 4$ at $K=16$, §6.4). The regimes of the reference study become corners of this arithmetic: serial with small B favours the single copy (ISM); serial with large B favours the linearised majoriser (FISTA), whose work $\Theta(BMN)$ carries no r factor at all; abundant workers favour the full split (consensus in parallel). The interior covers the cases between the corners: with $W < B$ workers or a memory budget below $2BMN$, group size $r \approx B/W$ gives full worker utilisation at the smallest exact-solve overhead, and the σ law removes the tuning penalty that previously made switching methods a re-tuning exercise.

The channel dimension enters through the duality: under $\mathcal{R}_1$, C multiplies right-hand sides only (choose any grayscale cell; costs scale by C in the FFT and projection terms only), while under $\mathcal{R}_2$, C multiplies the block count and hence the rank. For colour ($C = 3$), image-grouping is the sweet spot — exact rank-3 solves, grayscale-consensus memory, K -way parallelism — which is precisely the new cell validated at scale in §6.3. As C grows (multi/hyperspectral), rank- C groups steepen as $\Theta(BCMN) = \Theta(KC^2MN)$ and the full split or the linearised majoriser take over, consistent with the FISTA advantage reported for hyperspectral CDL in [13]; transposing roles, the same arithmetic applied to the sparse-coding stage (where C plays K ’s part) explains that stage’s known preferences, though abstracting that stage is beyond this paper’s object.

8 What the template does not absorb

Honesty about the abstraction’s edges. (i) *Adaptive penalties and relaxation*. The reference solvers optionally adapt σ by residual balancing and over-relax with $\alpha = 1.8$; our runs pin both off, on both sides, matching the reference study’s own fixed-parameter comparisons. The template carries α ; residual-adaptive σ is a protocol layer that could sit on top of any cell but is not part of the derivation, and with adaptation active our bitwise agreements would relax to adaptation-path agreement. (ii) *Tiling is an embedding, not a cell*. The tiled method changes the spatial index set before the template applies; it lives one level below the partition parameter, and boundary effects of the concatenation are outside the per-frequency analysis. (iii) *Masked CDL*. The mask-decoupled updates add a second lifted block ($\mathbf{g}_1 = X\mathbf{d} - \mathbf{s}$ carrying $\frac{1}{2}\|W\mathbf{g}_1\|^2$); the master iteration extends in the natural way to two coupled splitting blocks — at the full split this recovers the structure of the hybrid mask-decoupling/consensus update of [4] — but we have not implemented the extension and claim nothing empirical about it. (iv) *The sparse-coding stage* is deliberately not abstracted: the study design holds it fixed, and the duality argument of §7 uses only its known structure. (v) *Scale*. The $\mathcal{R}_2$ colour runs were executed at $K = 20$ rather than the reference colour scale $K = 40$ for the memory reasons disclosed in §6.3; all other anchors are at the reference scales. (vi) *Interior tuning*. Our $\sigma \propto r$ law equalises convergence across partitions at the consensus endpoint’s constant; we did not grid-search the interior for absolute optimality, and the fixed- σ fan in Figure 2 shows what mis-scaled penalties cost.

9 Related work

The dictionary-update algorithms unified here were introduced across several papers: frequency-domain CDL by Bristow et al. [21], the equality-constraint ADMM update with CG and ISM solvers by Wohlberg [3], tiling, consensus, and the 3D method by Šorel and Šroubek [6], the FISTA update and the comparative analysis — including the 3D/consensus correspondence and the penalty-scaling arguments this paper generalises — by Garcia-Cardona and Wohlberg [4], with the subproblem-coupling structure from [9] and consensus ADMM from [7], whose “general form consensus” is the generic ancestor of our grouped partitions. Our contribution relative to these is the single parameterised derivation, the machine-precision demonstration that it *is* the shipped solvers, and the interior and channel cells it opens. The published multi-channel extension of this line, Garcia-Cardona and Wohlberg [13], carries the existing updates to C channels, reduces the shared-dictionary problem to CK folded images, contributes a product (convolutional $\times$ spectral) dictionary model and robust-FISTA variants, and derives C -scaling laws for algorithm parameters; it mentions channel-wise consensus only as an unevaluated possibility for the *sparse-coding* stage (its footnote 3). None of the grouped-partition cells, the channel-frequency form, or Proposition 1 appear there, and our lattice locates their product-dictionary model as a different extension direction (a factored constraint set, not a partition choice). Also adjacent: accelerated proximal-gradient consensus for CDL [14] (a linearised full-split cell with acceleration, marked in Table 3), block-proximal-gradient CDL with designed majorisers by Chun and Fessler [15] (MM applied to the whole biconvex problem rather than inside the update’s splitting), slice-based CDL [16], and multivariate/rank-one channel models for M/EEG [17], whose rank-constrained channel coupling would be a constraint-set cell rather than a partition cell. Linearised and majorised augmented-Lagrangian schemes are classical [18, 19, 20]; the template is their specialisation to (2) with the partition and basis structure made explicit and exploited.

10 Limitations and conclusion

Beyond the edges listed in §8: our validation is on the reference protocol’s image class and λ ; the machine-precision agreements are protocol-pinned (deterministic solvers, fixed penalties) and would soften to statistical agreement under adaptive settings; and the regime arithmetic of §7 was verified in serial timings, with the parallel claims resting on the synchronisation structure rather than on a distributed implementation. The grouped and channel-lifted cells were validated for convergence and cost, not proposed as universally superior — their value is that the lattice makes such cells available, costed, and tuned ($\sigma \propto r$) without new derivations.

The broader point is methodological. A body of separately-derived algorithms for one subproblem admitted a single two-line problem statement (3) and one iteration (4)–(7) that reproduces each of them exactly, predicts their parameter scalings, exposes an equivalence that had only been observed along one axis, and yields practical new members in its interior. The lattice, not any single cell, is the result: it replaces “which of these algorithms should I implement?” with a small amount of arithmetic over (B, r, M, N, C, W) inside an implementation one already has.

References

- [1] M. S. Lewicki and T. J. Sejnowski. Coding time-varying signals using sparse, shift-invariant representations. In *Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 11, pp. 730–736, 1999.
- [2] M. D. Zeiler, D. Krishnan, G. W. Taylor, and R. Fergus. Deconvolutional networks. In *Proc. IEEE CVPR*, pp. 2528–2535, 2010.
- [3] B. Wohlberg. Efficient algorithms for convolutional sparse representations. *IEEE Trans. Image Process.*, 25(1):301–315, 2016.
- [4] C. Garcia-Cardona and B. Wohlberg. Convolutional dictionary learning: A comparative review and new algorithms. *IEEE Trans. Computational Imaging*, 4(3):366–381, 2018.
- [5] B. Wohlberg. Efficient convolutional sparse coding. In *Proc. IEEE ICASSP*, pp. 7173–7177, 2014.
- [6] M. Šorel and F. Šroubek. Fast convolutional sparse coding using matrix inversion lemma. *Digital Signal Processing*, 55:44–51, 2016.
- [7] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein. Distributed optimization and statistical learning via the alternating direction method of multipliers. *Foundations and Trends in Machine Learning*, 3(1):1–122, 2010.
- [8] A. Beck and M. Teboulle. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM J. Imaging Sci.*, 2(1):183–202, 2009.
- [9] C. Garcia-Cardona and B. Wohlberg. Subproblem coupling in convolutional dictionary learning. In *Proc. IEEE ICIP*, pp. 1697–1701, 2017.
- [10] B. Wohlberg. SPORCO: A Python package for standard and convolutional sparse representations. In *Proc. 15th Python in Science Conference*, pp. 1–8, 2017.
- [11] B. Wohlberg. Convolutional sparse representation of color images. In *Proc. IEEE Southwest Symp. Image Anal. Interp. (SSIAI)*, pp. 57–60, 2016.
- [12] M. J. Huiskes, B. Thomee, and M. S. Lew. New trends and ideas in visual concept detection: The MIR Flickr retrieval evaluation initiative. In *Proc. Int. Conf. Multimedia Information Retrieval*, pp. 527–536, 2010.
- [13] C. Garcia-Cardona and B. Wohlberg. Convolutional dictionary learning for multi-channel signals. In *Proc. 52nd Asilomar Conf. Signals, Systems, and Computers*, pp. 335–342, 2018.
- [14] G. Silva and P. Rodríguez. Efficient algorithm for convolutional dictionary learning via accelerated proximal gradient consensus. In *Proc. IEEE ICIP*, pp. 432–436, 2018.
- [15] I. Y. Chun and J. A. Fessler. Convolutional dictionary learning: Acceleration and convergence. *IEEE Trans. Image Process.*, 27(4):1697–1712, 2018.
- [16] V. Pappyan, Y. Romano, J. Sulam, and M. Elad. Convolutional dictionary learning via local processing. In *Proc. IEEE ICCV*, pp. 5306–5314, 2017.
- [17] T. Dupré la Tour, T. Moreau, M. Jas, and A. Gramfort. Multivariate convolutional sparse coding for electromagnetic brain signals. In *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 31, pp. 3295–3305, 2018.
- [18] N. Parikh and S. Boyd. Proximal algorithms. *Foundations and Trends in Optimization*, 1(3):127–239, 2014.
- [19] D. R. Hunter and K. Lange. A tutorial on MM algorithms. *The American Statistician*, 58(1):30–37, 2004.
- [20] Y. Sun, P. Babu, and D. P. Palomar. Majorization-minimization algorithms in signal processing, communications, and machine learning. *IEEE Trans. Signal Process.*, 65(3):794–816, 2017.
- [21] H. Bristow, A. Eriksson, and S. Lucey. Fast convolutional sparse coding. In *Proc. IEEE CVPR*, pp. 391–398, 2013.